In [3]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.tree import DecisionTreeClassifier
from sklearn.tree import DecisionTreeClassifier
```

In [4]:
```
df = pd.read_csv("IRIS.csv")
df
```

Out[4]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

In [5]:
```
X = df[['sepal_length', 'sepal_width', 'petal_length', 'petal_width']]
Y = df['species']
```

In [6]:
```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3, random_
```

In [7]:
```
dtree = DecisionTreeClassifier(criterion='gini', max_depth=3, random_state=42)
dtree.fit(X_train, Y_train)
```

Out[7]:

▾ DecisionTreeClassifier

▶ Parameters

In [8]:
```
Y_pred = dtree.predict(X_test)
```

In [9]:
```
print("Accuracy:", accuracy_score(Y_test, Y_pred))
print("\nCongusion Matrix:\n", confusion_matrix(Y_test, Y_pred))
```

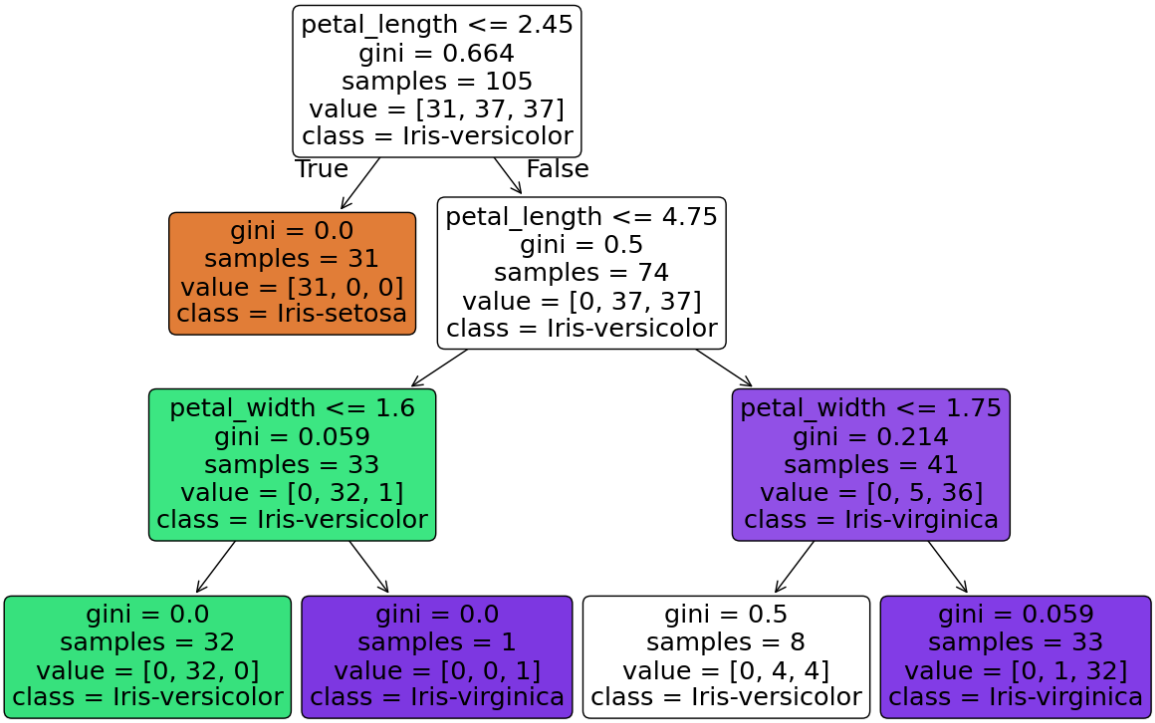
```
Accuracy: 1.0

Congusion Matrix:
 [[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

In [10]:
```
import matplotlib.pyplot as plt
from sklearn.tree import plot_tree

plt.figure(figsize=(15 ,10))

plot_tree(dtree,
          feature_names=X.columns,
          class_names=dtree.classes_,
          filled=True,
          rounded=True
)
plt.show()
```



In [ ]:

In [ ]:

In [ ]: